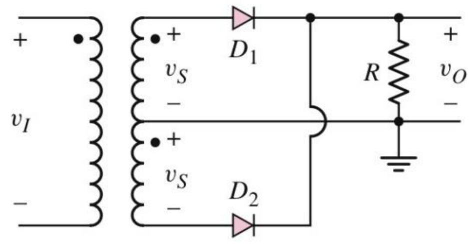
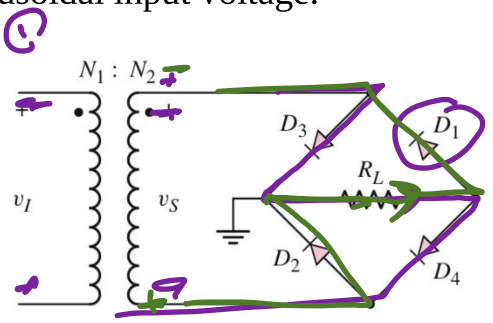
ECE204a _2nd year_1st semester_chap 02

1. Draw the output waveform for the following circuit (Full-Wave Centre-Tapped Transformer Rectification) with a sinusoidal input voltage.

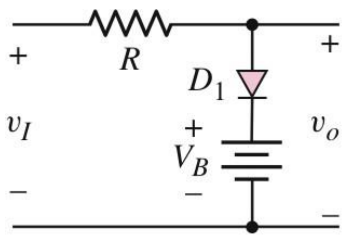


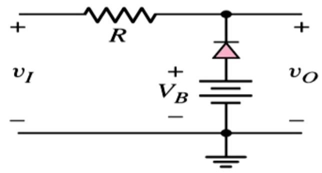
2. Draw the output waveform for the following circuit (Full-Wave Bridge Rectifier) with a sinusoidal input voltage.



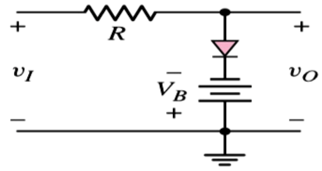
3. Draw the output waveform for the following circuit (parallel-based Clipper Circuits) with a sinusoidal input voltage.

1.

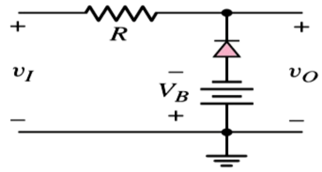




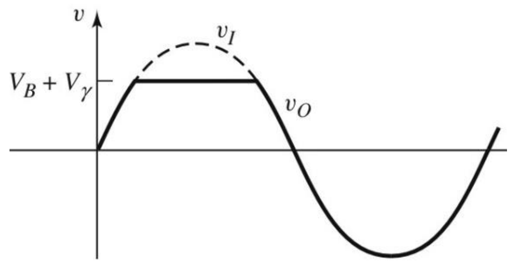
R
+
v_I
−
+
V_B
−
+
v_O
−



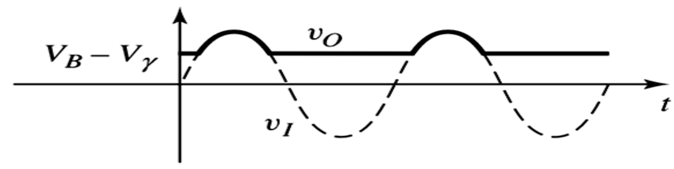
R
+
v_I
−
−
V_B
+
+
v_O
−



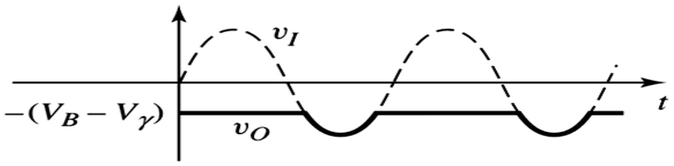
R
+
v_I
−
−
V_B
+
+
v_O
−



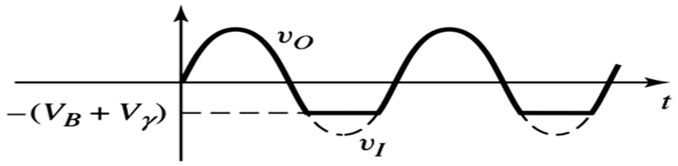
v
v_I
$V_B + V_\gamma$
v_O



(a)

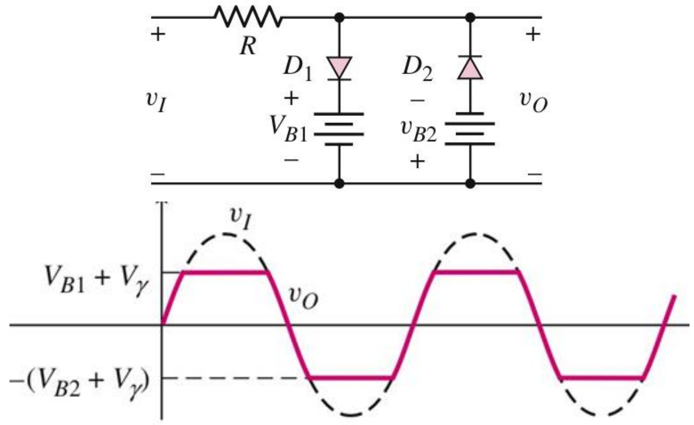


(b)

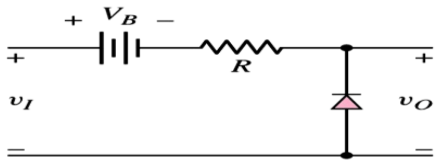


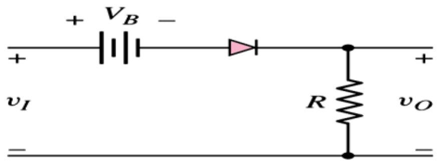
(c)

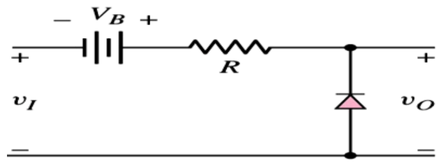
4. Draw the output waveform for the following circuit (Positive and negative clipping circuits) with a sinusoidal input voltage.

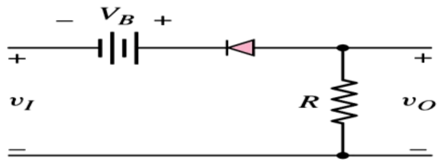


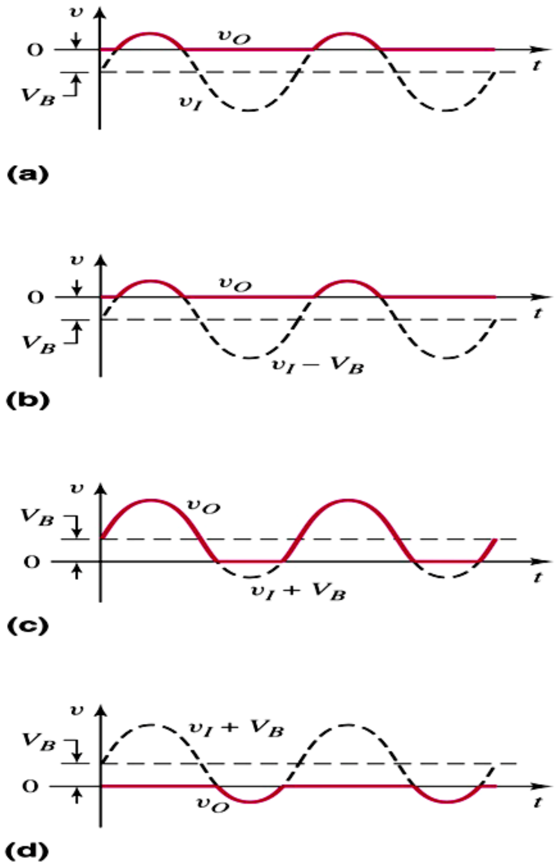
5 Draw the output waveform for the following circuit (series-based Clipper Circuits) with a sinusoidal input voltage


+ V_B −
R
+
v_I
−
+
v_O
−


+ V_B −
+
v_I
−
R
+
v_O
−


− V_B +
R
+
v_I
−
+
v_O
−


− V_B +
+
v_I
−
R
+
v_O
−



ECE204a _2nd year_1st semester_chap 03

1. Calculate the current in an n-channel enhancement-mode MOSFET with the following parameters: $V_{TN} = 0.4$V, W = 20 µm, L = 0.8 µm, $\mu_n = 650\ \text{cm}^2/\text{V–s}$, $t_{ox} = 200$ Å, and $\epsilon_{ox} = (3.9)(8.85 \times 10^{-14})$ F/cm. The transistor is biased in the saturation region with:

(a) $V_{GS} = 0.8$ V and (b) $V_{GS} = 1.6$ V

Solution: Saturation region

$i_D = K_n(v_{GS} - V_{TN})^2$ and $K_n = \frac{W}{L} \cdot \frac{k_n'}{2}$ or $K_n = \frac{W}{L} \cdot \frac{\mu_n C_{ox}}{2}$

$$K_n = \frac{W\mu_n\epsilon_{ox}}{2Lt_{ox}}$$

$$K_n = \frac{W\mu_n\epsilon_{ox}}{2Lt_{ox}} = \frac{(20 \times 10^{-4})(650)(3.9)(8.85 \times 10^{-14})}{2(0.8 \times 10^{-4})(200 \times 10^{-8})}$$

$K_n = 1.40\ \text{mA/V}^2$

(a) **V_{GS}=0.8V**

$i_D = K_n(v_{GS} - V_{TN})^2 = (1.40)(0.8 - 0.4)^2 = 0.224\text{mA}$

(b) **V_{GS}= 1.6V**

$i_D = (1.40)(1.6 - 0.4)^2 = 2.02\text{mA}$

2. Determine the source to drain voltage required to bias a p-channel enhancement mode MOFET in the saturation region.

Consider K_p= 0.2m A/v^2, V_{TP} = -0.5 V, and i_D = 0.50 mA.

Solution:

The drain current in the saturating region given as:

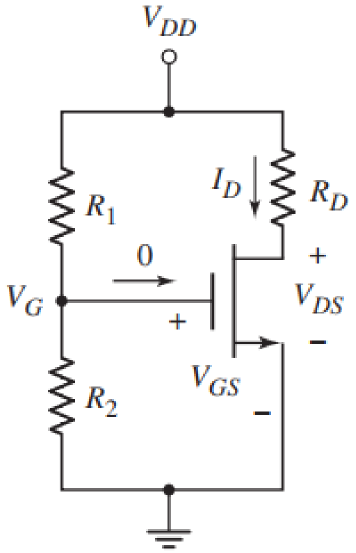
$i_D = K_p(v_{SG} + V_{TP})^2$

0.50= 02 (v_{SG}-0.5)^2 then V_{SG}= 2.08 V

The voltage is required to bias the p-channel MOSFET in the saturation region:

$v_{SD} > v_{SD}(\text{sat}) = v_{SG} + V_{TP} = 2.08 - 0.5 = 1.58\text{V}$

3. Calculate the **drain current** and **drain-to-source voltage** for a common-source circuit using **n-channel MOSFET enhancement mode** circuit as shown below which is the same as figure 3.1. Find **power dissipated** in the transistor, where R_1 = 30 kΩ, R_2 = 20 kΩ, R_D = 20 kΩ, V_{DD} = 5V, V_{TN} = 1V, and K_n = 0.1 mA/V^2.



Solution:

$$V_G = V_{GS} = \left(\frac{R_2}{R_1 + R_2}\right)V_{DD} = \left(\frac{20}{20+30}\right)(5) = 2\,\mathrm{V}$$

The procedure to obtain the drain current is to assume the transistor is in the saturation region and then compare the V_{DS} value with the V_{DS} (sat):

- **Drain current**

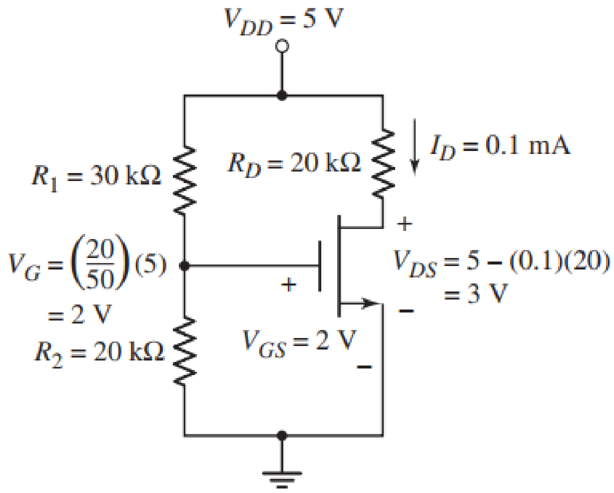
$I_D = Kn\,(V_{GS} - V_{TN})^2 = (0.1)\,(2 - 1)^2 = 0.1$ mA

- **Drain to source voltage**

Thus, the drain of the voltage source is obtained as follows:

$V_{DS} = V_{DD} - I_D\,R_D$

$V_{DS} = 5 - (0.1) \times (20) = 3$V



Since V_{DS} = 3 V > V_{DS} (sat) = $V_{GS} - V_{TN}$ = 2 − 1 = 1V, the transistor is already biased in the saturation region and our analysis is correct.

- **The power dissipated:**

$P_T = I_D V_{DS}$ = (0.1) x (3) = 0.3 mW

4. <u>Design</u> a biased circuit with a p-channel MOSFET with both positive and negative voltages. The specification specifies I_{DQ} = 100 μA, V_{DSQ} = 3 V, and V_{RS} = 0.8 V. The value of the larger bias resistor, either R_1 or R_2, is to be 200 kΩ. The transistor parameters k_P= 100 μA/V^2 and V_{TP} = -0.4 V used in the design circuit.

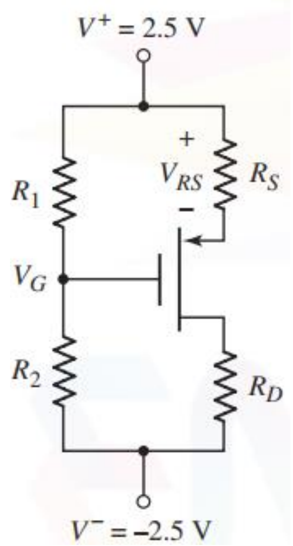


Figure 3.4 circuit configuration

Solution: Assume the transistor is biased in the saturation region

$I_D = K_p\,(V_{SG} + V_{TP})^2$ $\Rightarrow (V_{SG} + V_{TP})^2 = \sqrt{\frac{I_D}{k_p}}$

$$V_{SG} = \sqrt{\frac{I_{DQ}}{Kp}} + V_{TP} = \sqrt{\frac{100}{100}} - (-0.4) = 1.4\text{ V}$$

V_{SDQ} (sat) = $V_{SGQ} + V_{TP}$ = 1.4 - 0.4 = 1V

V_{SDQ} = 3 V > V_{SDQ} (sat) so the transistor is biased in the saturation region.

V_s

$V_G = V^+ - V_{RS} - V_{SG}$ = 2.5 - 0.8 - 1.4 = 0.3 V

With $V_G > 0$, the resistor R_2 will be the larger of the two bias resistors, so set **R_2 = 200kΩ**. The current through R_2 is then

$$I_{Bias} = \frac{V_G - V^-}{R_2} = \frac{0.3 - (-2.5)}{200} = 0.014\text{mA}$$

Regarding MOSFET since the current through R_1 is the same, we can find the value of R_1 to be.

$$\mathbf{R_1} = \frac{V^+ - V_G}{I_{DQ}} = \frac{2.5 - 0.3}{0.014} = 157\text{ k}\Omega$$

The source resistor value is found from.

$\mathbf{R_S} = V_{RS}/I_{DQ}$ = 0.8/0.1 = 8 kΩ

The voltage at the drain terminal equal to:

$V_D = V^+ - V_{RS} - V_{SD}$ = 2.5 - 0.8 - 3 = -1.3 V

Then the drain resistor value is found as

$$\mathbf{R_D} = \frac{V_D \ - V^-}{I_{DQ}} = \frac{-1.3-(-2.5)}{0.1} = 12\ \mathrm{k\Omega}$$

5. Determine the operation region and obtain the value of R_D for the below MOSFET circuit with a constant current supply. The quiescent values are $I_{DQ} = 250\ \mu A$ and $V_D = 2.5$ V and the transistor with $V_{TN} = 0.8$ V, kn' = 80 $\mu A/V^2$, and W/L=3 available.

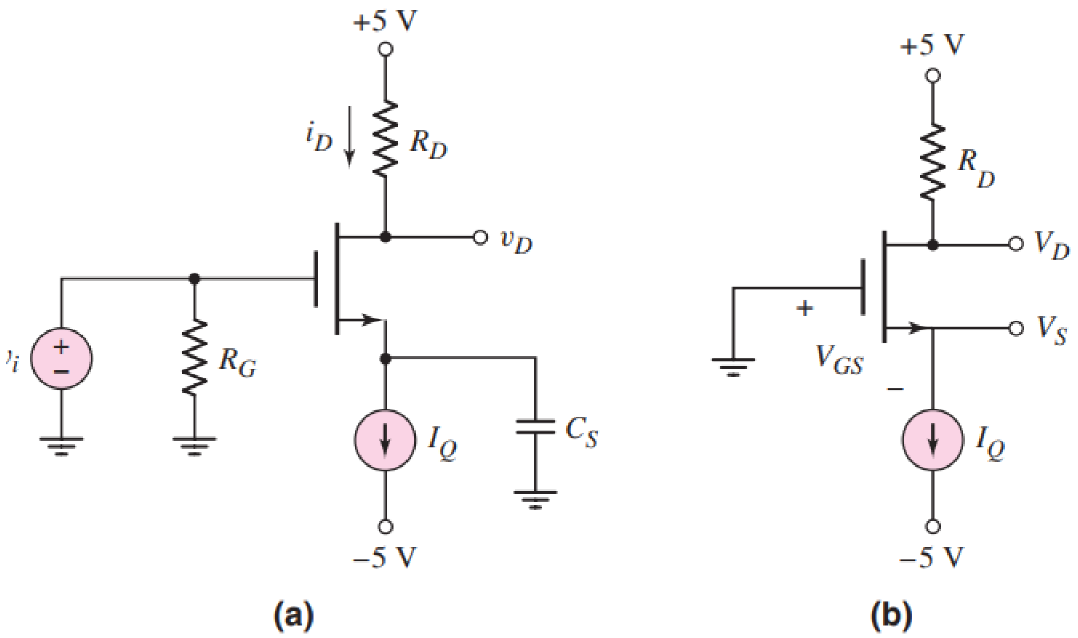


Figure 3.1 (a) NMOS common-source circuit biased with a constant-current source and (b) equivalent dc circuit

Solution:

$I_Q = I_{DQ} = 250\mu A$, and assuming the transistor is biased in the saturation region,

$$I_D = \frac{k_n'}{2} \cdot \frac{W}{L}(V_{GS} - V_{TN})^2$$

$$250 = \left(\frac{80}{2}\right) \cdot (3)(V_{GS} - 0.8)^2$$

$V_{GS} = 2.24$ V

The voltage at the source terminal is $V_S = -V_{GS} = -2.24$ V.

The drain current can also be written as

$$I_D = \frac{5 - V_D}{R_D}$$

For $V_D = 2.5$ V, we have

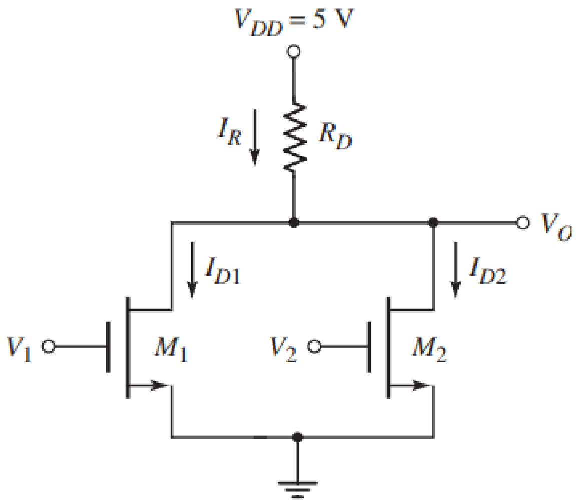
$$R_D = \frac{5 - 2.5}{0.25} = 10\text{k}\Omega$$

The drain-to-source voltage is

$V_{DS} = V_D - V_S = 2.5 - (-2.24) = 4.74$ V

Note: Since $V_{DS} = 4.74\text{ V} > V_{DS}\text{(sat)} = V_{GS} - V_{TN} = 2.24 - 0.8 = 1.44$ V, the transistor is biased in the saturation region, as initially assumed.

6. Determine the currents and voltages in a digital logic gate, for various input conditions. Consider the circuit shown in Figure below with circuit and transistor parameters $R_D = 20$ kΩ, $K_n = 0.1$ mA/V^2, $V_{TN} =$ 0.8V.



Solution:

Table 3.2 NMOS NOR logic circuit response

V_1(V)	V_2(V)	V_O(V)
0	0	High
5	0	Low
0	5	Low
5	5	Low

1. $V1 = V2 = 0$, both M1 and M2 are cut off and $V_O = V_{DD} = 5$ V.
2. $V_1 = 5$ V and $V_2 = 0$, transistor M1 is biased in the nonsaturation region, and then

$$I_R = I_{D1} = \frac{5 - V_O}{R_D} = K_n[2(V_1 - V_{TN})V_O - V_O^2]$$

Solving for the output voltage V_O, we obtain $V_O = 0.29$ V.

The currents are

$$I_R = I_{D1} = \frac{5 - 0.29}{20} = 0.236 \text{ mA}$$

3. For $V_1 = 0$ and $V_2 = 5$ V, we have $V_O = 0.29$ V and $I_R = I_{D2} = 0.236$ mA.
4. When both inputs rise to $V_1 = V_2 = 5$ V, we have $I_R = I_{D1} + I_{D2}$, or

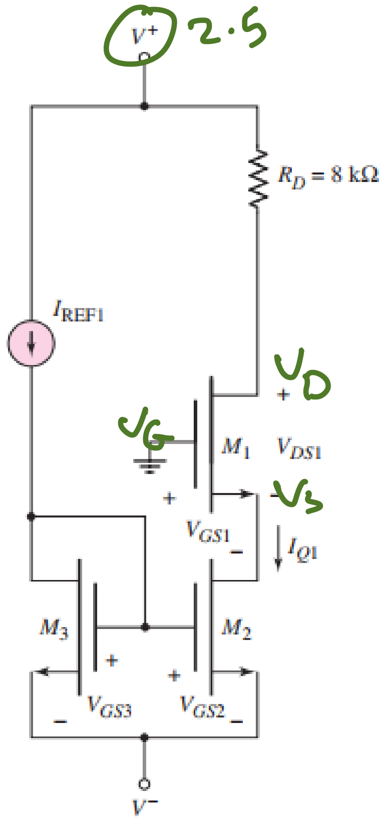
$$\frac{5 - V_O}{R_D} = K_n[2(V_1 - V_{TN})V_O - V_O^2] + K_n[2(V_2 - V_{TN})V_O - V_O^2]$$

Then solved for V_O to yield $V_O = 0.147$ V.

$$I_R = \frac{5 - 0.147}{20} = 0.243 \text{ mA}$$

$$I_{D1} = I_{D2} = \frac{I_R}{2} = 0.121 \text{ mA}$$

7. Determine the bias current I_{Q1}, the gate-to-source voltages for the transistors, and the drain-to-source voltages for M_1 for the circuit shown in Figure below. Assume the circuit parameters for $I_{REF1} = 200$ µA, $V^+ = 2.5$ V, and $V^- = -2.5$ V. Assume all the transistor parameters $V_{TN} = 0.4$V, $K_{n1} = 0.25$ mA /V^2, and $K_{n2} = K_{n3} = 0.15$ mA/V^2.



$V_{GS} = V_G - V_S$

$V_{GS} = 0 - V_S$

Solution:

The drain current in M_3 is $I_{D3} = I_{REF1} = 200\ \mu A$ and is given by the relation $I_{D3} = K_{n3}(V_{GS3} - V_{TN})^2$ (the transistor is biased in the saturation region).

Solving for the gate-to-source voltage, we find

$$V_{GS3} = \sqrt{\frac{I_{D3}}{K_{n3}}} + V_{TN} = \sqrt{\frac{0.2}{0.15}} + 0.4$$

$V_{GS3} = 1.555$ V

We note that $V_{GS3} = V_{GS2} = 1.555$ V. We can write

$I_{D2} = I_{Q1} = K_{n2}(V_{GS2} - V_{TN})^2 = 0.15(1.555 - 0.4)^2$

or $I_{Q1} = 200\ \mu A$

The gate-to-source voltage V_{GS1} (assuming M_1 is biased in the saturation region) can be written as

$$V_{GS1} = \sqrt{\frac{I_{Q1}}{K_{n1}}} + V_{TN} = \sqrt{\frac{0.2}{0.25}} + 0.4$$

Or V_{GS1} = 1.29 V

The drain-to-source voltage is found from as $V_{GS1} = V_{G1} - V_{S1} = -V_{S1}$ and $V_{DS1} = V_{D1} - V_{S1}$ then

$V_{DS1} = V^{+} - I_{Q1} R_D - (-V_{GS1})$

= 2.5 - (0.2) (8) - (-1.29)

or V_{DS1} = 2.19 V

We may note that M_1 is indeed biased in the saturation region

Note: Since the current mirror transistors M_2 and M_3 are matched (identical parameters) and since the gate-to-source voltages are the same in the two transistors, the bias current, I_{Q1}, is equal to (i.e., mirrors) the reference current, I_{REF1}.

ECE204a _2nd year_1st semester_chap 04

1. Calculate the transconductance of an n-channel MOSFET with parameters V_{TN} = 0.4 V, k_n' = 100µA/V^2, and W/L = 25. Assume the drain current is I_D = 0.40 mA.

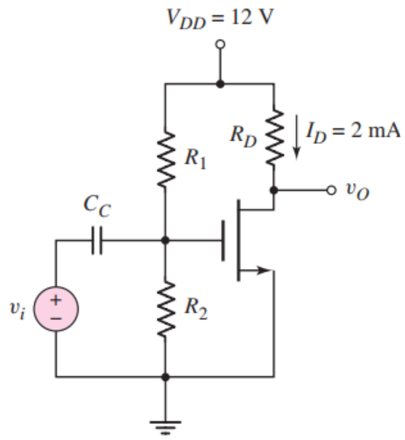
Solution: The conduction parameter is

$$K_n = \frac{k_n'}{2} \cdot \frac{W}{I} = \left(\frac{0.1}{2}\right)(25) = 1.25\text{mA/V}^2$$

Assuming the transistor is biased in the saturation region, the transconductance is determined as

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(1.25)(0.4)} = 1.41 \text{ mA/V}$$

2. Determine the small signal voltage gain and input and output resistors for the common source amplifier circuit shown in Figure below, the transistor parameters are: V_{DD} = 3.3 V, R_D = 10 KΩ, R1 = 140 KΩ, R2 = 60 KΩ, and R_{Si} = 4 KΩ. Transistor parameters where: V_{TN} = 0.4 V, K_n = 0.5 mA/V^2, and λ = 0.02 V^{-1}.



Solution:

Step 1: The dc or quiescent gate-to-source voltage

$$V_G = \left(\frac{R_2}{R_1+R_2}\right)(V_{DD}) = \left(\frac{60}{140+60}\right)(3.3) = 0.99 \text{ V} = V_{GSQ}$$

Since $V_{GS}=V_G-V_S$ and $V_S=0$ then $V_{GS}=V_G$, the quiescent drain current is

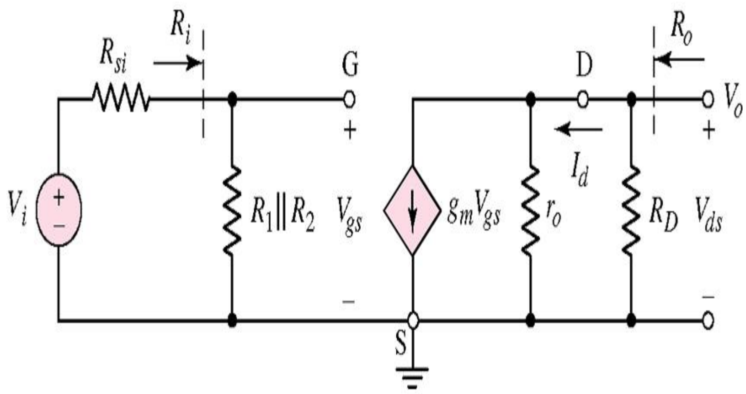
$$I_{DQ} = K_n\left(V_{GSQ} - V_{TN}\right)^2 = (0.5)(0.99 - 0.4)^2 = 0.174 \text{ mA}$$

The quiescent drain-to-source voltage is

$$V_{DSQ} = V_{DD} - I_{DQ}R_D = 3.3 - (0.174)(10) = 1.56 \text{ V}$$

Since $V_{DSQ} > V_{GSQ} - V_{TN}$, the transistor is biased in the saturation region.

Step 2: AC analysis is the small-signal analysis



The small-signal transconductance g_m is then

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(0.5)(0.174)} = 0.590 \text{ mA/V}$$

The output resistance is

$$r_o = \frac{1}{\lambda I_Q} = \frac{1}{(0.02)(0.174)} = 287\text{k}\Omega$$

The input resistance to the amplifier is

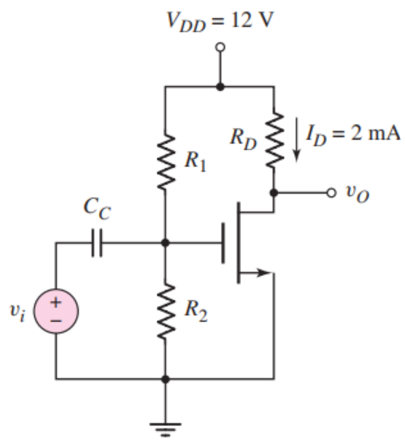
$$R_i = R_1 \| R_2 = 140\|60 = 42\text{k}\Omega$$

Voltage Gain: The small-signal voltage gain is

Av = -gm(r_o//R_D) $\left(\frac{R_\text{i}}{R_\text{i}+R_\text{si}}\right)$ = -(0.59) (28710 $\left(\frac{42}{42+4}\right)$

- Av = -5.21
- The input resistance to the amplifier is $Ri = R1//R2$.
- The output resistance is therefore $Ro = R_D//r_o$.

3. Design and determine the gain of the small signal voltage for the circuit shown in Figure below. Let R1// R2 = 100 KΩ. The circuit is design with Q-point located in the middle of the saturation region so that I_{DQ} = 2 mA and the transistor parameters V_{TN} = 1V, k_n'= 80 µA/V^2, W/L = 25, and λ = 0.015 V^{-1}.



Solution:

Step 1 DC analysis

Since the Q-point is to be in the middle of the saturation region, so the current at the transition point must be 4 mA.

The conductivity parameter

$$K_n = \frac{k'_n}{2} \cdot \frac{W}{I} = \left(\frac{0.080}{2}\right)(25) = 1mA/V^2$$

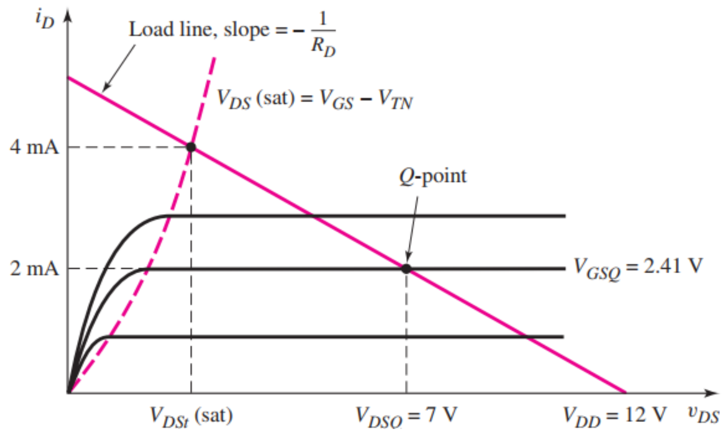
The value of VDS (sat) at the transition point first determine the V_{GSt}

$I_{Dt} = 4 = K_n (V_{GSt} - V_{TN})^2 = 1(V_{GSt} - 1)^2$

$V_{GSt} = 3V$

$V_{DSt} = V_{GSt} - V_{TN} = 3 - 1 = 2V$

Since Q-point is in the middle of the saturation region, then $V_{DSQ} = 7V$, which would yield a 10 V peak-to-peak symmetrical output voltage.



$$V_{DSQ} = V_{DD} - I_{DQ}R_D$$

$$R_D = \frac{V_{DD} - V_{DSQ}}{I_{DQ}} = \frac{12 - 7}{2} = 2.5k\Omega$$

We can determine the required quiescent gate-to-source voltage from the current equation, as follows:

$$I_{DQ} = 2 = K_n\left(V_{GSQ} - V_{TN}\right)^2 = (1)\left(V_{GSQ} - 1\right)^2$$

$V_{GSQ} = 2.41$ V= V_G as V_S=0

$$V_{GSQ} \quad = 2.41 = \left(\frac{R_2}{R_1 + R_2}\right)(V_{DD}) = \left(\frac{1}{R_1}\right)\left(\frac{R_1 R_2}{R_1 + R_2}\right)(V_{DD})$$

$$= \frac{R_i}{R_1} \cdot V_{DD} = \frac{(100)(12)}{R_1}$$

$R_1 = 498k\Omega$ and $R_2 = 125k\Omega$

Step 2 (ac analysis): The small-signal transistor parameters are

$$g_m = 2\sqrt{K_n I_{DQ}} = 2\sqrt{(1)(2)} = 2.83\ \text{mA/V}$$

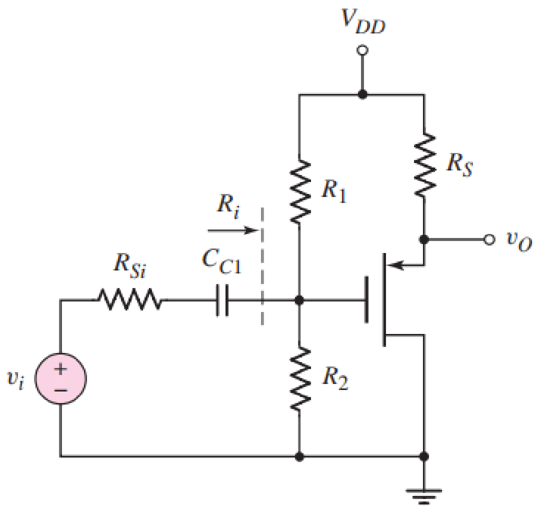
$$r_o = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.015)(2)} = 33.3\text{k}\Omega$$

The small signal voltage gain is

$$A_v = \frac{V_o}{V_i} = -g_m(r_o \parallel R_D) = -(2.83)(33.3 \parallel 2.5)$$

$$A_v = -6.58$$

4. Design a follower source amplifier and find the W/L ratio for P-channel enhancement mode MOSFET for the circuit shown in Figure below with the circuit parameters V_{DD} = 20V and R_{SI} = 4kΩ. Q-point values at the centre of the load line with I_{DQ} = 2.5 mA. The input resistance is R_i = 200 kΩ. The transistor with small signal voltage gain is Av = 0.90 with parameters $V_{TP} = -2$ V, k'_p = 40 μA/V².



Solution:

<u>(DC Analysis):</u> KVL equation about the source-to-drain loop

$V_{DD} = V_{SDQ} + I_{DQ} R_S$

20 = 10 + (2.5) R_s.

R_S = 4 k.

$$I_{DQ} = K_p\left(V_{GSQ} + V_{TP}\right)^2$$

$$2.5 = 0.784\left(V_{SGQ} - 2\right)^2$$

Then a quiescent source-to-gate voltage of $V_{SGQ} = 3.79$ V.

The quiescent source-to-gate voltage can also be written as

$$V_{SGQ} = \left(V_{DD} - I_{DQ}R_S\right) - \left(\frac{R_2}{R_1 + R_2}\right)(V_{DD})$$

$$\left(\frac{R_2}{R_1 + R_2}\right) = \left(\frac{1}{R_1}\right)\left(\frac{R_1 R_2}{R_1 + R_2}\right) = \left(\frac{1}{R_1}\right) \cdot R_i$$

$$3.79 = [20 - (2.5)(4)] - \left(\frac{1}{R_1}\right)(200)(20)$$

The bias resistor R_1 is then found to be

$R_1 = 644\text{k}\Omega$

Since $R_i = R_1 \parallel R_2 = 200\text{k}\Omega$, then

$R_2 = 290\text{k}\Omega$

AC design: The small signal voltage gain of this circuit is the same as that

Follow the source to the NMOS device.

$$A_v = \frac{V_o}{V_i} = \frac{g_m R_S}{1 + g_m R_S} \cdot \frac{R_i}{R_i + R_{Si}}$$

$$0.90 = \frac{g_m(4)}{1 + g_m(4)} \cdot \frac{200}{200 + 4}$$

The transconductance $g_m = 2.80$ mA/V.

The transconductance can be written as

$$g_m = 2\sqrt{K_p I_{DQ}}$$

$$2.80 \times 10^{-3} = 2\sqrt{K_p(2.5 \times 10^{-3})}$$

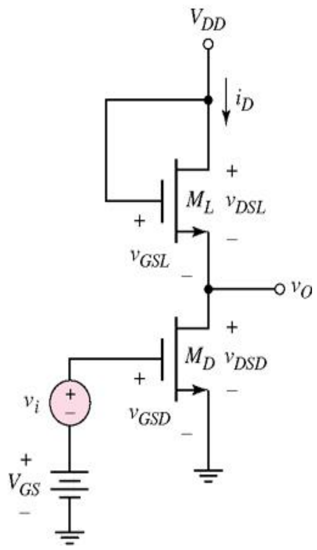
$$K_p = 0.784 \times 10^{-3}\ \text{A/V}^2$$

The conduction parameter, as a function of width-to-length ratio, is

$$K_p = 0.784 \times 10^{-3} = \frac{k_p'}{2} \cdot \frac{W}{L} = \left(\frac{40 \times 10^{-6}}{2}\right) \cdot \left(\frac{W}{L}\right)$$

$$\frac{W}{L} = 39.2$$

5. Find the $(W/L)_D$ of NMOS amplifier with an enhancement load as in Figure below with a small signal voltage gain of $|Av| = 10$. The circuit is biased at $V_{DD} = 5$ V, the transistors parameters are $V_{TN} = 1$V, $k'_{nD} = k'_{nL}$, $\lambda = 0$ and $(W/L)_L = 1$.



Solution:

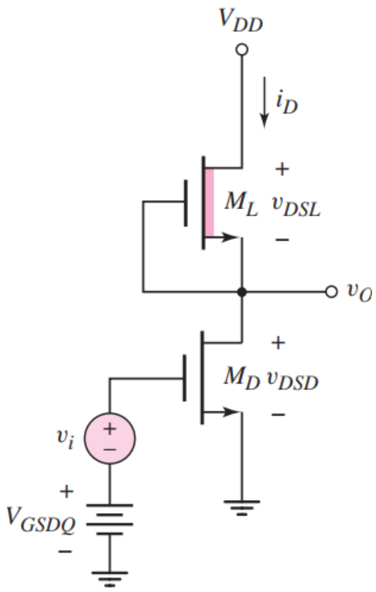
$$\approx -\frac{g_{mD}}{g_{mL}} = -\sqrt{\frac{K_{nD}}{K_{nL}}} = -\sqrt{\frac{(W/L)_D}{(W/L)_L}}$$

$$|A_v| = 10 = \sqrt{\frac{(W/L)_D}{(W/L)_L}}$$

$$\left(\frac{W}{L}\right)_D = 100\left(\frac{W}{L}\right)_L$$

Then $(W/L)_D = 100$

6. Determine the small signal voltage gain of an NMOS amplifier with depletion load for the circuit shown in the figure below, assume the transistor parameters $V_{TND} = +0.8$ V, $V_{TNL} = -1.5$ V, $K_{nD} = 1$ mA/V^2, $K_{nL} = 0.2$ mA/V^2, and $\lambda_D = \lambda_L = 0.01$ V^{-1} with $I_{DQ} = 0.2$ mA.



Solution:

$$g_{mD} = 2\sqrt{K_{nD} I_{DQ}} = 2\sqrt{(1)(0.2)} = 0.894 \text{ mA/V},.$$

Since $\lambda_D = \lambda_L$,

$$r_{oD} = r_{oL} = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.01)(0.2)} = 500\,\text{k}\Omega$$

$$A_v = -g_{mD}(r_{oD} \| r_{oL}) = -(0.894)(500 \| 500) = -224$$

7. Determine the small signal voltage gain of the CMOS amplifier for the circuit shown in the figure below. Assume transistor parameters V_{TN} = +0.8 V, V_{TP} = −0.8 V, k'_n = 80 μA/V^2, k'_p = 40 μA/V^2, (W/L) n = 15, (W/L) p = 30, and $\lambda_D = \lambda_L$ = = 0.01 V^{-1}. Also, assume I_{Bias} = 0.2 mA.

Solution:

$$g_{mn} = 2\sqrt{K_n I_{DQ}} = 2\sqrt{\left(\frac{k'_n}{2}\right)\left(\frac{W}{L}\right)_n I_{\text{Bias}}}$$

$$= 2\sqrt{\left(\frac{0.08}{2}\right)(15)(0.2)} = 0.693 \text{ mA/V}$$

Since $\lambda_n = \lambda_p$ then the output resistances are

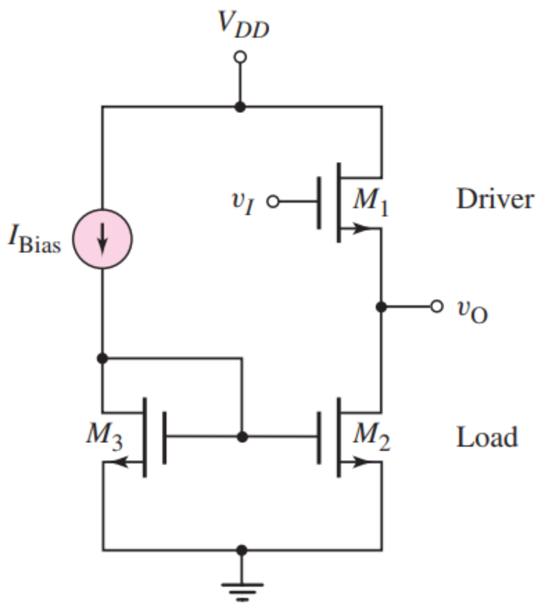
$$r_{\text{on}} = r_{\text{op}} = \frac{1}{\lambda I_{DQ}} = \frac{1}{(0.01)(0.2)} = 500\,\text{k}\Omega$$

The small signal voltage gain is then obtained

$$A_v = -g_m(r_{\text{on}} \| r_{\text{op}}) = -(0.693)(500\|500) = -173$$

CMOS Source-Follower Amplifier

8. Determine the small signal voltage gain and output impedance of the source amplifier shown in the figure below. Assume the reference bias current is $I_{Bias} = 0.20$ mA and the bias voltage is $V_{DD} = 3.3$ V. Assume that all transistors are identical with the parameters $V_{TN} = 0.4$V, $K_n = 0.20$ mA/V^2, and $\lambda = 0.01$ V^{-1}.



Solution:

$$A_v = \frac{V_o}{V_i} = \frac{g_{m1}(r_{o1} \| r_{o2})}{1 + g_{m1}(r_{o1} \| r_{o2})}$$

$$g_{m1} = 2\sqrt{K_n I_{D1}} = 2\sqrt{(0.20)(0.20)} = 0.40\,\text{mA/V}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda I_D} = \frac{1}{(0.01)(0.20)} = 500\,\text{k}\Omega$$

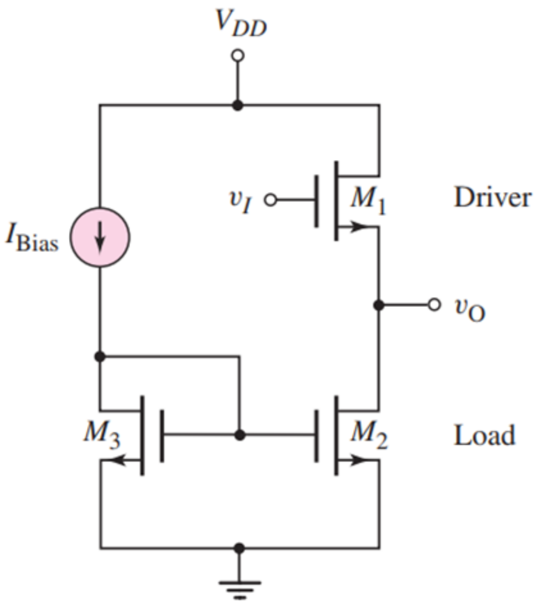
$$A_v = \frac{(0.40)(500\|500)}{1 + (0.40)(500\|500)}$$

Av = 0.990

$$R_o = \frac{1}{0.40} \|500\| \, 500$$

Ro = 2.48 kΩ

9. Determine the small-signal voltage gain and output resistance of the common-gate circuit shown in Figure below. Assume the reference bias current is I_{Bias} = 0.20 mA and the bias voltage is V_{DD} = 3.3 V. Assume that the transistor parameters are V_{TN} = +0.4 V, V_{TP} =−0.4 V, K_n = 0.20 mA/V^2, K_p = 0.20 mA/V^2, and $\lambda n = \lambda p$ = 0.01 V^{-1}.
M2 and M 3 are matched transistors and have the same source-to-gate voltage, the bias current in M1 is $I_{D1} = I_{Bias}$ = 0.20 mA.



Solution:

$$A_v = \frac{\left(g_{m1} + \frac{1}{r_{o1}}\right)}{\left(\frac{1}{r_{o2}} + \frac{1}{r_{o1}}\right)}$$

$$g_{m1} = 2\sqrt{K_n I_{D1}} = 2\sqrt{(0.20)(0.20)} = 0.40\ \text{mA/V}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda I_{D1}} = \frac{1}{(0.01)(0.20)} = 500\ \text{k}\Omega$$

$$A_v = \frac{\left(0.40 + \frac{1}{500}\right)}{\left(\frac{1}{500} + \frac{1}{500}\right)}$$

Av = 101

$$R_o = \frac{V_x}{I_x} = r_{o1} \| r_{o2} = 500 \| 500$$

Ro = 250 k